

Supplementary

1 Progression Analysis with Cognitive States

To assess progression across cognitive statuses, we evaluate the difference in ground truth and predictions of three subjects with similar previous time-point (82 years) and follow-up (84 years) ages from different cognitive stages (Figure 1). AD-DAE shows consistently lower errors than other methods, with error magnitude increasing with disease severity due to larger anatomical changes. AD-DAE preserves subject-specific anatomy while introducing progression-correlated changes. SITGAN shows higher errors in progression-related regions despite preserving subject identity. BrLP and DE-CVAE produce spatially diffuse errors with lesser localization to progression-relevant structures, while GAN-based unpaired methods (CAAE, IPGAN, UVCGAN) do not capture the progression correlating to the cognitive categories. Overall, AD-DAE achieves *lower* errors and models progression *consistent* with *cognitive status variation*.

2 How Well Is 3D Information Encoded in the Latent Space of the DAE?

Figure 2 illustrates the organization of axial slices from 3D volumes at different locations (70^{th} , 85^{th} , and 105^{th}) in the latent representational space of the DAE. Across the embedding space, axial slices originating from the same 3D volume are spatially co-located, indicating consistent encoding of volumetric information.

For each volume, the latent-space distances between the 70^{th} and 85^{th} slices, and between the 70^{th} and 105^{th} slices, are visualized using solid and dashed lines, respectively. Distances corresponding to axially closer slices (solid lines) are relatively smaller than those between axially distant slices (dashed lines). This relationship is observed across disease categories (CN, MCI, and AD, shown in blue, orange, and green) as well as across different age groups.

The spatial co-location of slices from the same volume, together with higher-level grouping by disease category and age, indicates that the latent representational space effectively encodes both subject-specific 3D structural information and progression-related characteristics. Overall, these results suggest that the

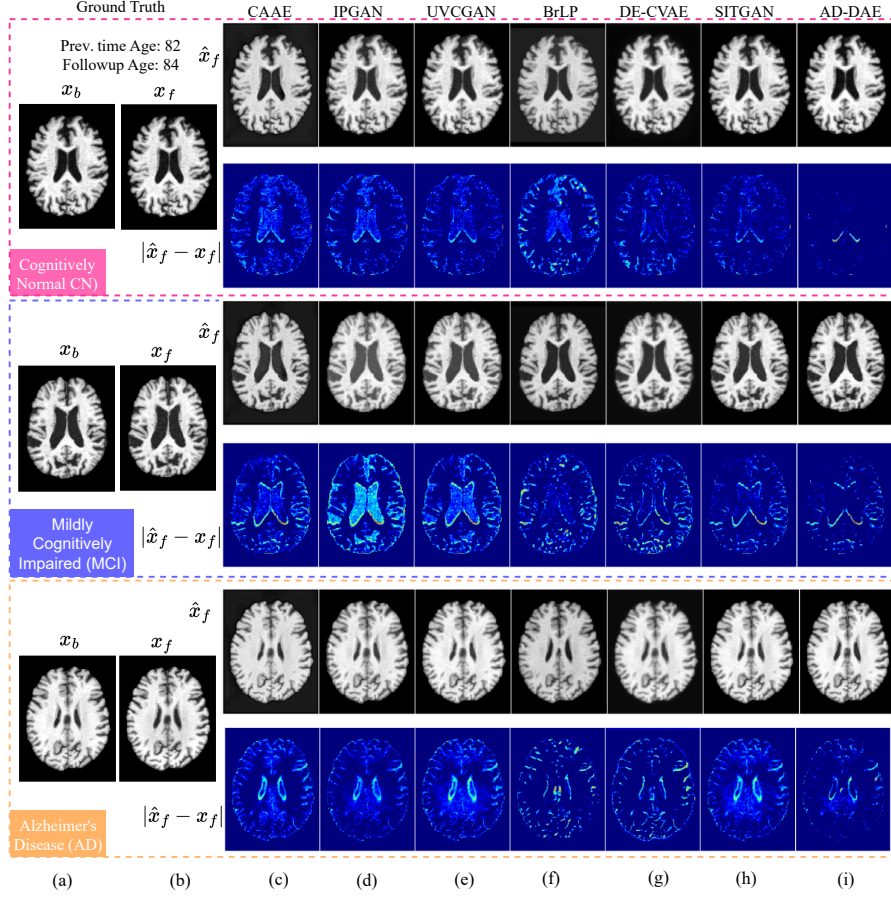


Figure 1: Top to bottom: Images and error maps for CN, MCI, and AD cohorts. Left to right: (a) Ground truth previous time-point (age 82), (b) Ground truth follow-ups (age 84), (c)–(i) Generated follow-up images and error maps (relative to ground truth follow-up) for AD-DAE and baseline methods. Overall, AD-DAE shows relatively lower errors in comparison to the other methods.

DAE latent space preserves volumetric coherence while capturing meaningful disease- and age-related variations.

3 Quantitative Evaluation of Progression–Identity Separation in the Latent Space

To quantitatively evaluate whether AD-DAE separates progression-related factors from subject-specific identity in the latent space, we perform a *latent swap*

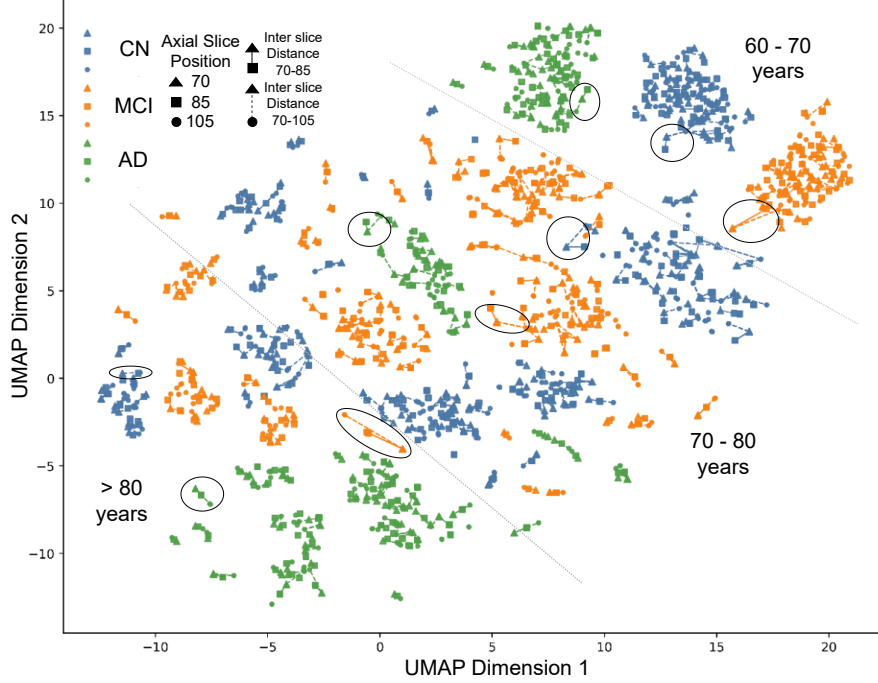


Figure 2: UMAP plot of the latent representations (z) of three axial sections from all the volumes of ADNI *Test Set*. The three slices- 70th, 85th, and 105th along the axial plane are denoted by triangle, square, and circle, respectively. The distances from the 70th to 85th and 70th to 105th are denoted by straight and dashed lines, respectively, for all the volumes. This plot represents that the spatial location of the 3D volumes are meaningfully embedded within the latent representational space.

experiment on the *Latent Swap Dataset*. In this experiment, the first m latent dimensions ($m = 50$), corresponding to progression-related components, are exchanged between paired subjects prior to image generation, while the remaining $(d - m)$ dimensions are preserved to retain subject identity.

For a paired CN subject x_b^B and AD subject x_b^A , progression-conditioned latent representations are computed as $z_f^B = \mathcal{E}(x_b^B) + [\mathcal{A}(v_d^B, v_a^B); \mathbf{0}]$ and $z_f^A = \mathcal{E}(x_b^A) + [\mathcal{A}(v_d^A, v_a^A); \mathbf{0}]$. Latent swapping is performed by constructing mixed representations $[z_f^B[0 : m]; z_f^A[m : d]]$ and $[z_f^A[0 : m]; z_f^B[m : d]]$, which are decoded to generate follow-up images $\hat{x}_f^A$ and $\hat{x}_f^B$, respectively.

Progression transfer is quantitatively assessed using normalized ventricular volumes. For AD $\rightarrow$ CN transformations with an age gap ≥ 2 years, $\hat{X}_f^A$ exhibits a reduced normalized ventricular volume of $V_{\hat{X}_f^A}^r/V_{\hat{X}_f^A} = 0.0105$, compared to

0.0115 for the original AD image $V_{X_b^A}^r/V_{X_b^A}$, indicating ventricular contraction consistent with CN progression. Conversely, for CN→AD transformations with an age gap < 2 years, $\hat{X}_f^B$ shows an increased normalized ventricular volume of 0.0119, relative to 0.0105 in X_b^B , reflecting ventricular enlargement characteristic of AD progression.

Preservation of subject identity is evaluated using image similarity metrics. The PSNR between $(\hat{x}_f^A, x_b^A)$ is 26.01 dB, and between $(\hat{x}_f^B, x_b^B)$ is 25.86 dB, indicating that identity-related anatomical structures are largely preserved despite the transfer of progression-related factors.

Together, these quantitative results demonstrate that AD-DAE enables controlled transfer of progression-specific anatomical changes while maintaining subject-specific identity, validating effective separation of progression and identity factors in the latent space.

4 Data Statistics of Test Set

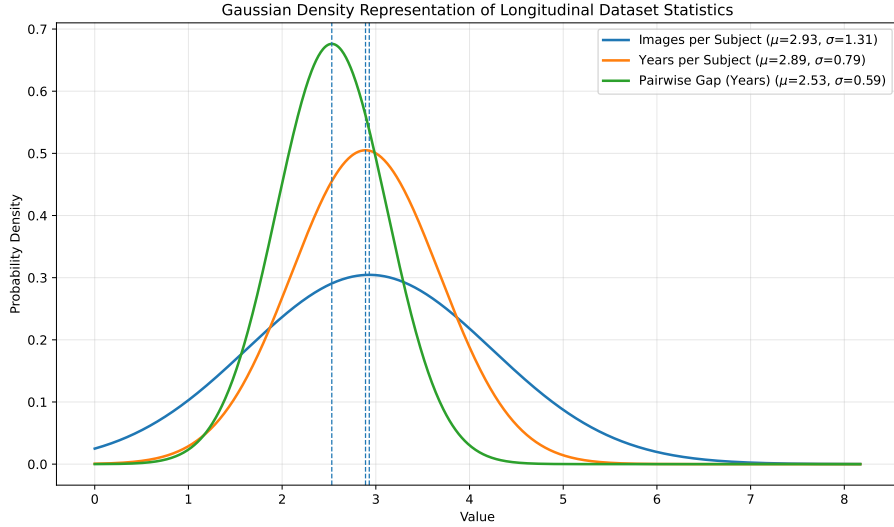


Figure 3: Gaussian density representation of the longitudinal dataset statistics. The distributions show the number of images per subject, longitudinal span per subject, and pairwise follow-up gap, summarized using their corresponding mean and standard deviation values.

In order to better represent the longitudinal statistics of the ADNI *Test Set*, a graphical representation of the data density is included in Figure 3. The plot summarizes the distribution of the number of images per subject, longitudinal span per subject, and pairwise follow-up gap using Gaussian density curves estimated from their mean and standard deviation values.

5 Ablation Studies for Regional Guidance

To validate the effect of the proposed regional guidance, we performed an ablation study by comparing AD-DAE with and without the approximate regional bounding-box guidance in the consistency module. The results show that incorporating regional guidance improves generation quality, increasing PSNR from 28.56 dB to 29.76 dB and reducing MSE from 0.0020 to 0.0017. This indicates that guiding the optimization toward AD-relevant anatomical regions improves the fidelity of generated progression images.

These improvements are attributed to the non-restrictive nature of the regional guidance. The guidance is implemented as an approximate regional bounding box, which encourages changes in the chosen primary AD-relevant regions while still allowing changes in secondary regions and maintaining anatomical consistency. To further enhance generation capabilities, this regional guidance can be extended to additional cortical regions associated with later-stage AD progression in future extensions of our work.